

NeuralGCM Hybrid Physics-AI for Weather and Climate

With Additional Tropical Cyclone Case Studies

EXECUTIVE SUMMARY

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Motivation

Weather forecasting matters because it supports decisions with direct social and economic consequences, from everyday choices to emergency preparations for extreme weather events [1, 2]. Recent data-driven forecasting systems have emerged from the long development of Numerical Weather Prediction (NWP), global observations, data assimilation, and reanalysis [3]. These systems have achieved strong benchmark accuracy while producing forecasts at much lower inference-time cost than current NWP models [4, 5]. NeuralGCM combines these two forecasting paradigms. It uses a physical dynamical core to represent large-scale atmospheric flow and a machine learning component to estimate unresolved sub-grid processes such as clouds, radiation and precipitation [6]. As one of the first hybrid models, NeuralGCM is a test case for whether physical modelling and ML can be reliably combined. Therefore, it requires rigorous evaluation to assess whether its reported performance is robust, physically reliable, and suitable for further development.

Methodology

The overarching aim of this project is to evaluate NeuralGCM as a weather and climate model. Following Murphy’s definition of forecast goodness [7], this means assessing not only benchmark accuracy, but also whether the model represents weather behaviour that is physically meaningful and useful for high-impact forecasting.

The project first reproduces key results from the original NeuralGCM paper [6]. This includes medium-range forecast accuracy metrics, the Hurricane Laura case study, and an annual tropical cyclone track experiment from a continuous NeuralGCM forecast. These experiments test whether the reported performance of NeuralGCM can be reproduced across benchmark metrics, a weather case study, and longer tropical cyclone behaviour.

The analysis is then extended to three additional 2021 tropical cyclone cases: Gulab and Shaheen, Faraji, and Seroja-Odette. These cases test whether NeuralGCM captures the broad path relative to IBTrACS, reproduces intensity measures and vertical coherence relative to ERA5, and captures each key dynamical challenge. Gulab and Shaheen test re-intensification, Faraji recurvature, and the Seroja-Odette two-storm interaction, where nearby cyclones rotate around a shared centre and one may absorb the other. This is known as the Fujiwhara effect [8].

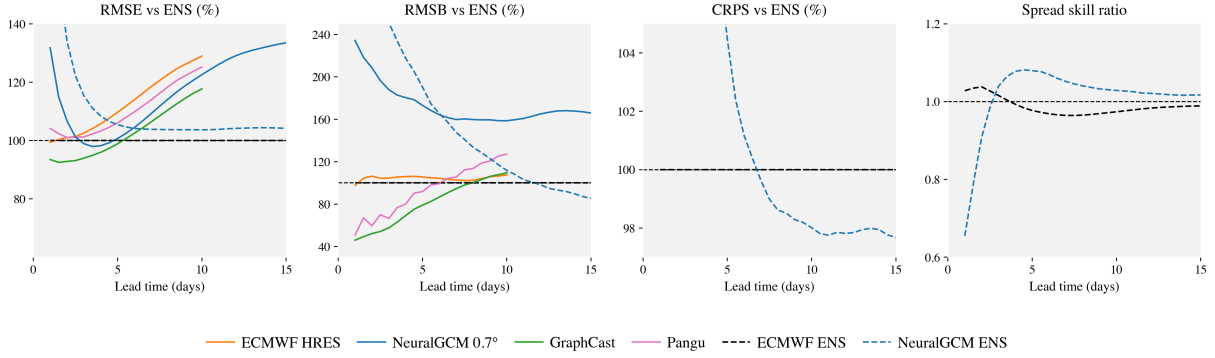


Figure 1: 500 hPa geopotential height reproduction diagnostics. The RMSE, RMSB and CRPS panels are shown as percentages of ECMWF ENS at the same lead time, so 100% indicates equal performance and values below 100% indicate lower error than ECMWF ENS. The final panel shows the spread skill ratio, where values close to 1 indicate better ensemble calibration.

Key Findings

The reproduction supports the overall conclusion of the original NeuralGCM paper, namely that NeuralGCM is competitive for medium-range weather forecasting. However, it is not an exact replication. The largest discrepancy is in the reproduced 500 hPa geopotential height results, which are weaker than in the original study. Figure 1 shows this result. The deterministic NeuralGCM 0.7° weakens relative to ECMWF ENS at longer lead times, with larger RMSE and RMSB, while the NeuralGCM ensemble shows a smaller probabilistic advantage in CRPS than reported in the paper. This may reflect differences in implementation details, including pre-processing and interpolation, which are not fully specified in the original paper. Other parts of the reproduction are more successful. The Hurricane Laura and annual tropical cyclone track experiments support qualitative reproduction, with NeuralGCM reproducing broad moisture structures in the Hurricane Laura case study and producing a realistic global distribution of annual tropical cyclone tracks.

The extension case studies give mixed answers to the three cyclone-focused aims. NeuralGCM often partly captures the tracks. The deterministic Faraji forecasts successfully reproduce the recurvature, but for Gulab and Shaheen and Seroja, the early tracks are reasonable before degrading once re-intensification or Fujiwhara interaction occurs. The ensemble tracks are broadly plausible for Gulab and

Shaheen, but perform poorly for Faraji, where they spread out and lose coherence. Across all case studies, NeuralGCM does not reliably reproduce intensity measures or vertical coherence. It produces weaker intensity measures and shallower vortex structures than ERA5, with the weakness strongest in the coarser and stochastic configurations. For Gulab and Shaheen, re-intensification is partly captured by NeuralGCM 0.7° , weaker in NeuralGCM 1.4° , and absent in the 1.4° ensemble. This suggests that NeuralGCM inherits some limitations seen in data-driven weather models, where tracks are often more reliable than cyclone intensity and structure [9, 10, 11].

The Seroja-Odette Fujiwhara case is the most novel extension result. Figure 2 shows that NeuralGCM 0.7° captures the binary vortex interaction most convincingly among the NeuralGCM configurations, although it remains weaker than ERA5. NeuralGCM 1.4° deterministic captures only a weakened interaction, while the 1.4° ensemble shows little to no coherent binary interaction. This suggests that NeuralGCM 0.7° can partly represent the Fujiwhara interaction, despite being much coarser than ERA5 0.25° , but that coarser resolution and stochastic member spread degrade the two-vortex structure. The result shows that NeuralGCM struggles when forecasts depend on maintaining two coherent interacting vortices.

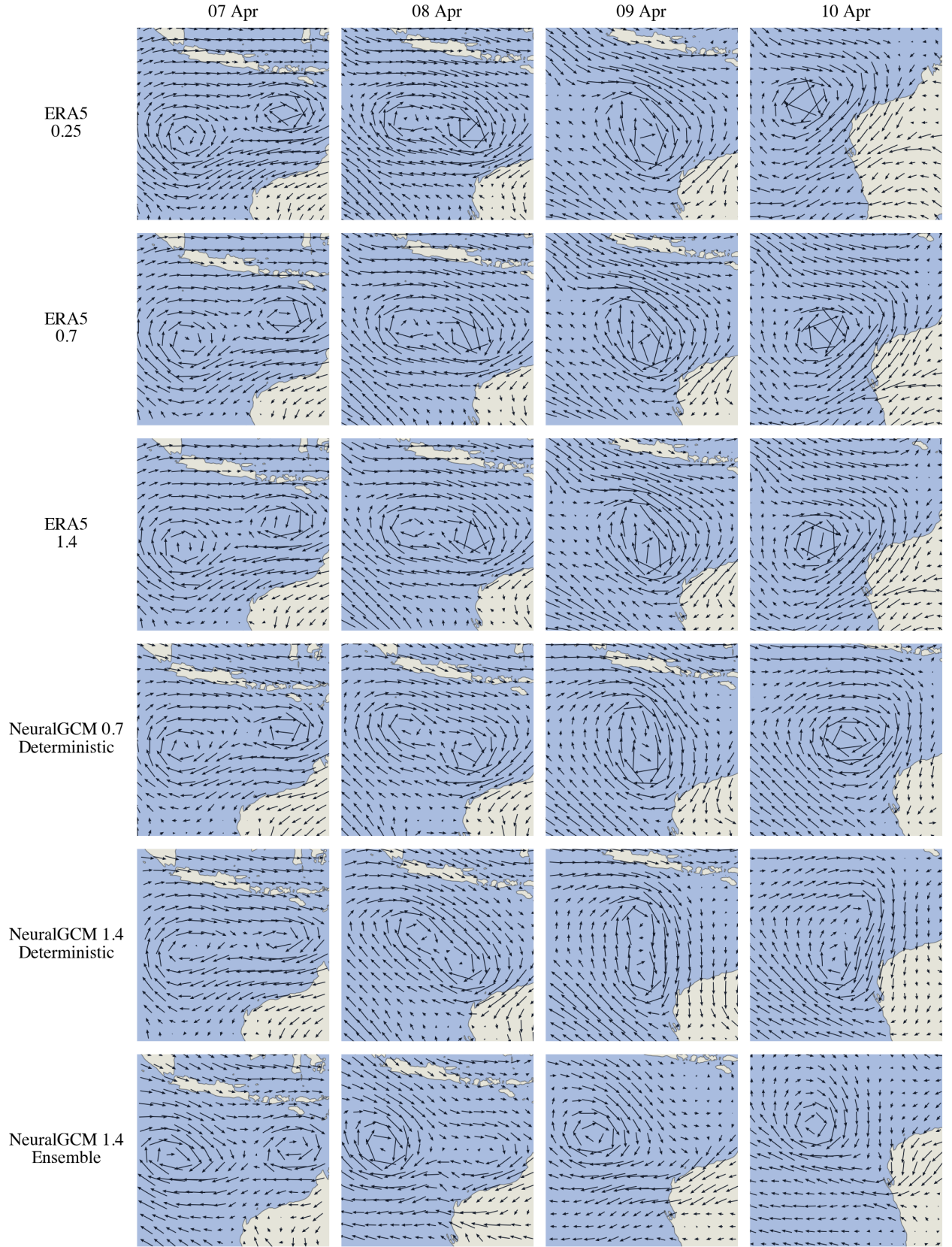


Figure 2: Midpoint-centred 850 hPa wind vectors for the Seroja-Odette Fujiwhara interaction on 7th to 10th April in ERA5 and NeuralGCM. Columns show the four dates and rows show ERA5 at 0.25°, 0.7°, and 1.4°, followed by NeuralGCM 0.7° deterministic, 1.4° deterministic, and 1.4° ensemble runs.

Research Impact and Conclusion

The first impact of this project is reproducibility. NeuralGCM is one of the first hybrid physics-AI models, so careful reproduction is needed before its reported performance can be treated as robust. This project broadly reproduces the original paper’s conclusions, but also shows that exact agreement is sensitive to implementation details such as pre-processing, interpolation and ensemble size.

The second contribution is a targeted evaluation of tropical cyclone dynamics, especially in Southern Hemisphere and dynamically complex cases. The Seroja-Odette case extends NeuralGCM evaluation to the Fujiwhara interaction, where two vortices interact and rotate around a common centre. To the best of the author’s knowledge, this behaviour has not previously been assessed in NeuralGCM or comparable data-driven weather models. The results show that NeuralGCM can represent some large-scale cyclone behaviour, but struggles with intensity, vertical coherence and rapid cyclone-scale structural changes.

Future work should test whether these weaknesses hold across a larger sample of storms, basins, lead times and model resolutions. More targeted diagnostics could investigate the winds that steer the cyclone, whether the same vortex is maintained through time, and the processes that strengthen or weaken 850 hPa cyclonic relative vorticity. NeuralGCM should also be assessed further as a climate model through longer forecasts that test model drift, stability and large-scale variability such as El Niño Southern Oscillation.

Overall, NeuralGCM shows promise as a hybrid physics-AI weather and climate model, but the results show that it still inherits some limitations associated with data-driven forecasting. It often captures the cyclone path, yet underestimates intensity measures and produces shallower vortex structures across model configurations.

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AI Declaration

Once a full draft was finished, AI was used for proofreading, minor wording improvements. Codex was used to assist with coding, as specified in the repository `README`.